Multi-Interval Report

Minh Quan Nguyen

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1 Model

Circuit and population. A circuit has N customers of class $j \in \{S, M, L\}$ with current I_j and load $P_j = VI_j$. Its composition $\boldsymbol{n} = (n_S, n_M, n_L)$, $\sum_j n_j = N$, is random:

$$\boldsymbol{n} \sim \text{Trinomial}(N; p_S, p_M, p_L), \qquad w(\boldsymbol{n}) = \frac{N!}{n_S!\, n_M!\, n_L!}\, p_S^{n_S} p_M^{n_M} p_L^{n_L}. \tag{1}$$

Intervals. A design has $K \in \{1, 2, 3\}$ free intervals $\mathcal{I}_k = [s_k, e_k)$ of 3 hours. Each medium and large customer is assigned one interval; inside it the customer draws I_j, outside it I_S. An allocation $\boldsymbol{a} = (m_1, \ldots, m_K, l_1, \ldots, l_K)$ gives the medium and large customers in each interval, with $\sum_k m_k = n_M$ and $\sum_k l_k = n_L$. The circuit current at time t is

$$A_{\boldsymbol{a}}(t) = NI_S + \sum_{k:\, t \in \mathcal{I}_k} \big((I_M - I_S)\, m_k + (I_L - I_S)\, l_k \big). \tag{2}$$

2 Objectives

Probability of overload. Each circuit composition $\boldsymbol{n}$ uses an allocation $\boldsymbol{a}(\boldsymbol{n})$, the same in every season. It overloads if its current exceeds the limit T at any time:

$$r(\boldsymbol{n}) = \begin{cases} 1, & \max_t A_{\boldsymbol{a}(\boldsymbol{n})}(t) > T, \\ 0, & \text{otherwise}, \end{cases} \qquad O_{\text{viol}} = \sum_{\boldsymbol{n}} w(\boldsymbol{n})\, r(\boldsymbol{n}). \tag{3}$$

Solar soak. A share u of Victoria's N_{VIC}/N circuits is on the plan (uptake). The plan load is the total load of these circuits, every customer's I_S plus the current the intervals add,

$$L(t) = u\, \frac{N_{\text{VIC}}}{N} \sum_{\boldsymbol{n}} w(\boldsymbol{n})\, V A_{\boldsymbol{a}(\boldsymbol{n})}(t). \tag{4}$$

Over $\mathcal{W} =$ 10:00–15:00, the same for every design, the soak is the rooftop solar the plan load soaks and the excess load is the plan load rooftop solar does not cover, both as a share of the rooftop solar in $\mathcal{W}$:

$$O_{\text{soak}} = \frac{\sum_{d \in \mathcal{D}} \sum_{t \in \mathcal{W}} \min\big(L(t), X_d(t)\big)}{\sum_{d \in \mathcal{D}} \sum_{t \in \mathcal{W}} X_d(t)}, \qquad E = \frac{\sum_{d \in \mathcal{D}} \sum_{t \in \mathcal{W}} \max\big(L(t) - X_d(t),\, 0\big)}{\sum_{d \in \mathcal{D}} \sum_{t \in \mathcal{W}} X_d(t)}, \tag{5}$$

where $X_d(t)$ is the Victorian rooftop solar generation on day d and $\mathcal{D}$ are the days of 2025.

3 Experimental Results

3.1 Setup

3.1.1 Parameters

Parameter	Value	Note
N	40	customers per circuit
T	1000 A	circuit limit
(I_S, I_M, I_L)	(20, 40, 100) A	current per customer class
V	240 V	
(p_S, p_M, p_L)	(0.69, 0.30, 0.01)	today's mix
N_{VIC}	2.9×10^6	Victorian customers (72,500 circuits)
u	13%	uptake: 9,425 circuits on the plan, calibrated (Section 3.1.3)
$X_d(t)$	2025 VIC1 rooftop solar	Open Electricity, 5-minute steps, UTC+10
Δt	5 min	rooftop values change every 30 minutes
Usage	All-On	full current for the whole interval

3.1.2 Interval choice

Each season s (summer: Dec–Feb, autumn: Mar–May, winter: Jun–Aug, spring: Sep–Nov) has its own intervals. They are chosen to cover the most rooftop solar: with $\bar{X}_s(t)$ the mean rooftop solar of season s at time t, the chosen set maximises

$$S_s(\mathcal{I}_1, \ldots, \mathcal{I}_K) = \sum_{k=1}^{K} \sum_{t \in \mathcal{I}_k} \bar{X}_s(t)\, \Delta t, \tag{6}$$

over 3-hour intervals that start every 0.5 hours and lie inside 08:00–16:00 (ties: the most rooftop solar covered by their union). Two intervals do not overlap; of three intervals, the first and last do not overlap and the middle one overlaps both. With one set per season, a circuit overloads if it overloads in any season, and soak uses each season's intervals on that season's days.

Season	1 interval	2 intervals	3 intervals
Summer	11:30–14:30	10:00–13:00, 13:00–16:00	10:00–13:00, 11:30–14:30, 13:00–16:00
Autumn	11:30–14:30	10:00–13:00, 13:00–16:00	10:00–13:00, 11:30–14:30, 13:00–16:00
Winter	11:00–14:00	10:00–13:00, 13:00–16:00	10:00–13:00, 11:00–14:00, 13:00–16:00
Spring	11:00–14:00	09:30–12:30, 12:30–15:30	09:30–12:30, 11:00–14:00, 12:30–15:30

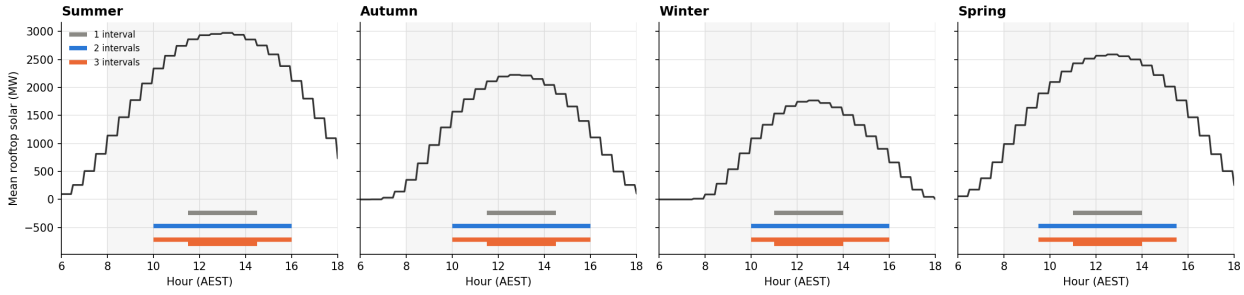


Figure 1: Mean rooftop solar per season and the chosen intervals. Shaded: 08:00–16:00.

3.1.3 Uptake

The plan load should be able to absorb the rooftop solar: it peaks at the midday peak of the average day of 2025, $\hat{X}$:

$$u = \frac{\hat{X}}{N_{\text{VIC}}\big(P_S + p_M(P_M - P_S) + p_L(P_L - P_S)\big)} = 12.8\%, \tag{7}$$

$$\hat{X} = \max_t \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} X_d(t) = 2{,}380 \text{ MW}. \tag{8}$$

The resulting plan load is in Figure 2.

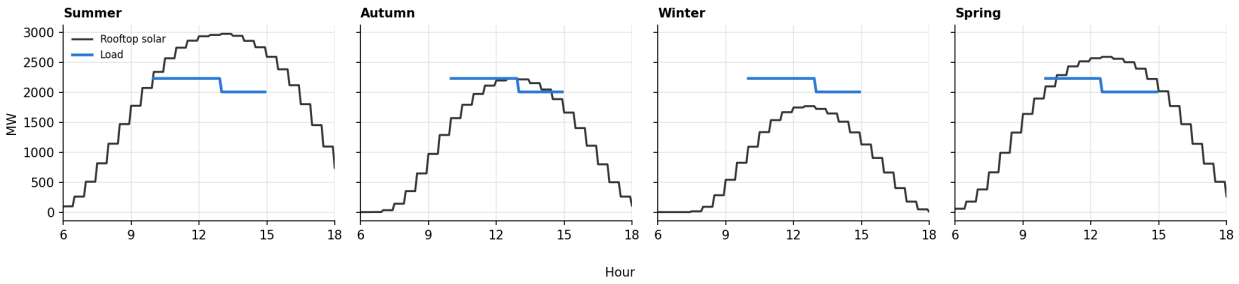


Figure 2: Plan load (two intervals) and mean rooftop solar.

3.2 Customer Mixes and Allocation

Other uptakes. Besides $u = 13\%$ we show $u = 10\%$ and 20%. The plan load grows in proportion (Figure 3).

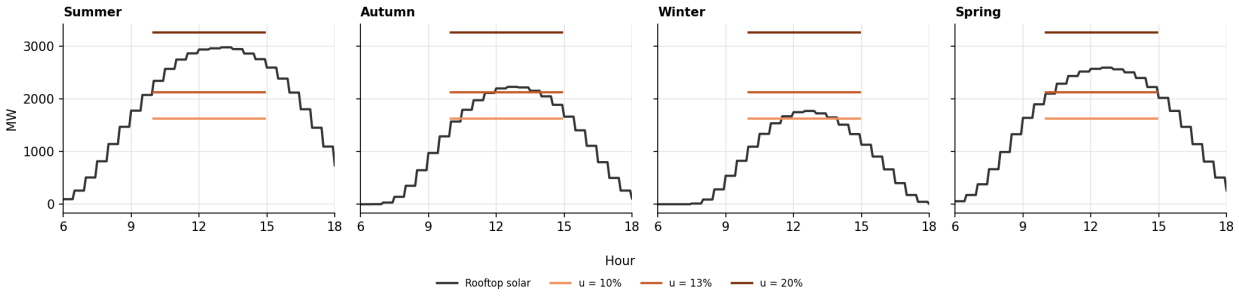


Figure 3: Plan load (two intervals, even split) for each uptake.

3.2.1 Allocation strategy

An allocation strategy decides how each circuit splits its medium and large customers between the intervals. We compare four kinds of strategy for each design:

- **Share rule:** every circuit uses the same split. Each share is a multiple of 5% (0%, 5%, …, 100%), which gives 21 possible rules for two intervals and 231 for three; all are evaluated.
- **Random:** every circuit composition draws its own split at random, uniformly over all splits; 300 such strategies per design.
- **Lowest overload:** every circuit composition gets its own split. If some split keeps the circuit under T, only those splits are used, otherwise the lowest-peak split; among these, the splits maximise O_{soak}.

- **Best soak:** every circuit composition gets its own split; the splits maximise O_{soak}, and among those O_{viol} is as low as possible.

The optimized point is the point that has the closest distance to the red point. The distance of a strategy from it is

$$\delta = \sqrt{O_{\text{viol}}^2 + \left(1 - O_{\text{soak}}\right)^2}, \tag{9}$$

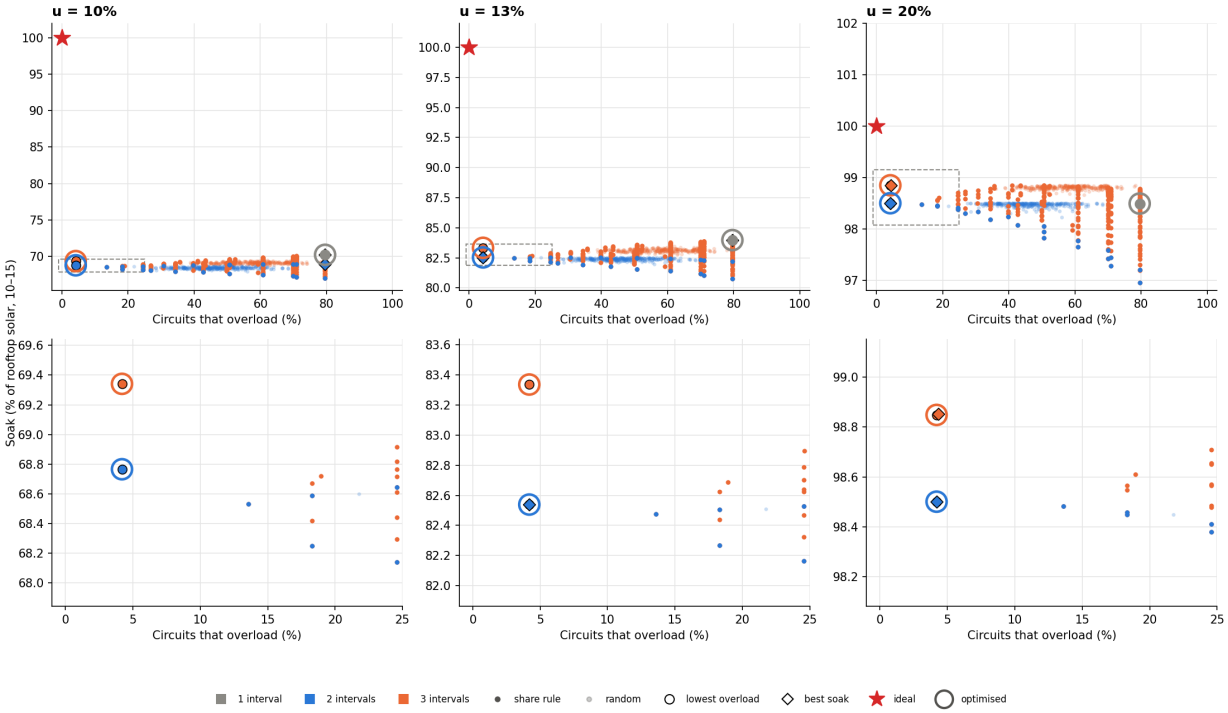


Figure 4: Every strategy at today's mix. Ring: optimised. Bottom: dashed box enlarged.

				O_{soak} (%)			
Design	Strategy	Split (%)	O_{viol}	$u = 10\%$	$u = 13\%$	$u = 20\%$	δ, $u = 13\%$
1 interval	–	100	79.6%	70.2	84.0	98.5	0.81
2 intervals	share rule	53/47	13.6%	68.5	82.5	98.5	0.22
	random	60/40	21.8%	68.6	82.5	98.4	0.28
	lowest overload	65/35	4.2%	68.8	82.5	98.5	0.18
	best soak	65/35	4.2%*	68.9	82.5	98.5	0.18
3 intervals	share rule	53/0/47	13.6%	68.5	82.5	98.5	0.22
	random	38/26/36	36.0%	69.0	83.0	98.8	0.40
	lowest overload	23/47/30	4.2%	69.3	83.3	98.8	0.17
	best soak	6/94/0	79.6%*	70.2	84.0	98.9	0.81

Split and O_{viol} at $u = 13\%$. *Two intervals: 79.6% at $u = 10\%$; three intervals: 4.3% at $u = 20\%$.

By season. Soak at $u = 13\%$, today's mix, lowest-overload strategy, as % of each season's rooftop solar between 10:00 and 15:00:

Design	Summer	Autumn	Winter	Spring	Year
1 interval	74.0	90.9	97.3	81.0	84.0
2 intervals	72.6	89.4	96.5	79.1	82.5
3 intervals	73.1	90.4	97.1	80.3	83.3

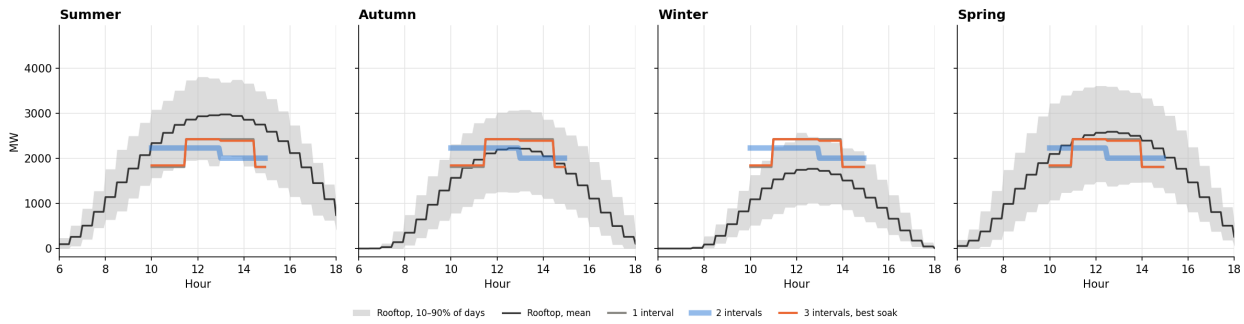


Figure 5: Plan load at the calibrated uptake and rooftop solar.

3.2.2 Customer mixes

Experiment. We vary the share of customers who add load in a free window, $\rho = p_M + p_L$, keeping medium and large customers in the ratio $30:1$:

$$p_M = \tfrac{30}{31}\rho, \qquad p_L = \tfrac{1}{31}\rho, \qquad p_S = 1-\rho, \qquad \rho \in \{0.31\, i/16 : i = 1, \ldots, 32\}, \tag{10}$$

so ρ runs from 1.9% to 62%; today's mix is $\rho = 31\%$. Each design traces one curve of $(O_{\text{viol}}, O_{\text{soak}})$.

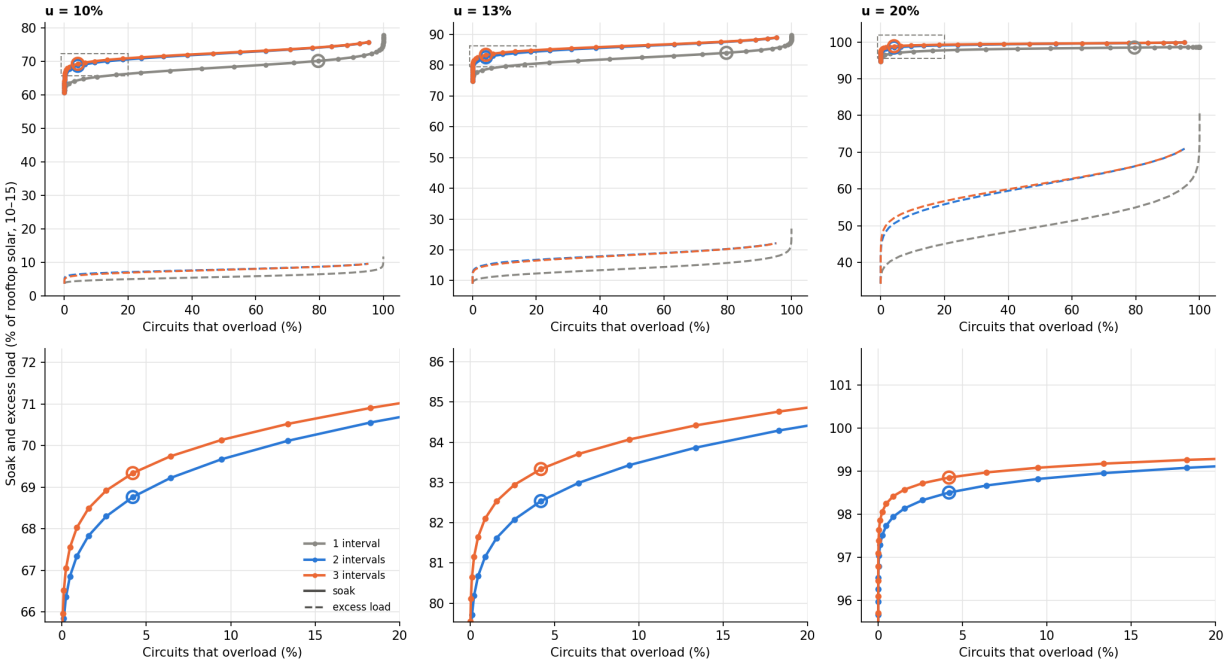


Figure 6: Overload against soak (solid) and excess load (dashed) as ρ grows. Circles: today's mix. Bottom: dashed box enlarged.

Probability of avoiding an upgrade. A circuit needs an upgrade if it overloads, so the probability that the scheme avoids one is $1 - O_{\text{viol}}$.

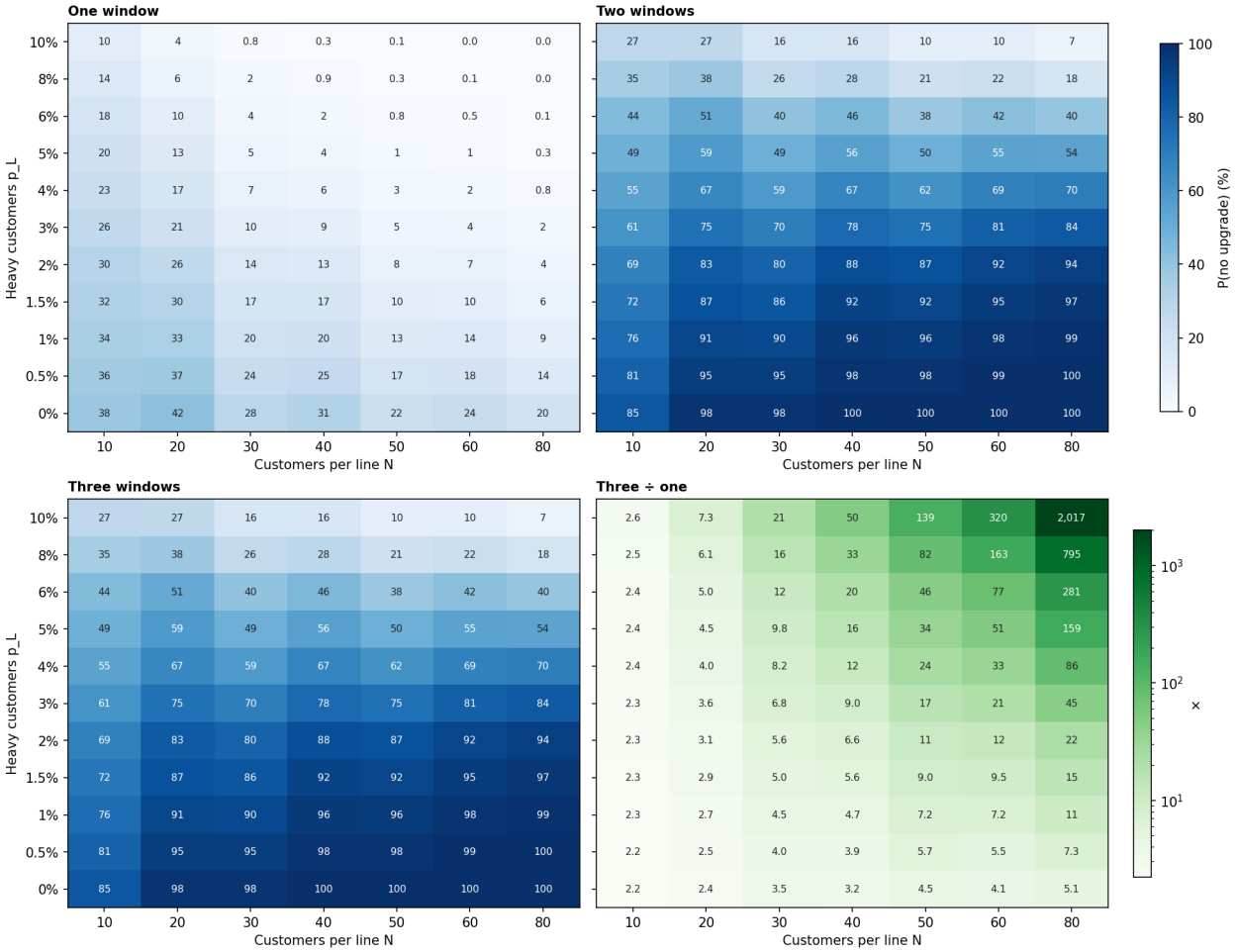


Figure 7: Probability of avoiding an upgrade. Green: three windows ÷ one.